Assignment 3 : Student Grade Sheet (GPA Calculator / Transcript)

Covers

Week 4 Classes & Objects, constructors, encapsulation, composition and Week 5 (up to <iomanip>)- getline + cin safely, error recovery, neat table formatting

Objectives

- Design two classes (Course, Student) with private fields, constructors, and const getters.
- Practice composition: Student contains many Courses.
- Handle mixed input correctly (**>>** with **getline, cin.clear, cin.ignore**).
- Build a formatted transcript using **<iomanip> (setw, left/right, fixed, setprecision)**.
- Compute weighted GPA from credit hours and grade points.

Submission Guideline

- Submit the **.h** and **.cpp** files on Brightspace.
- Include a short header comment with your name, date, and at the end of the code, write a 3–5 sentence reflection (what was tricky / what you learned).
- Make sure your code compiles and runs on an IDE without error.
- **DO NOT** submit a **.zip/.doc/.docx/.pdf** or any document files.

Grading Rubric (25 points total)

1.	Class design and composition (two classes):	**6 points**
2.	Input handling (getline/ >>, input validation, use of clear/ignore:	**4 points**
3.	Formatted output (iomanip):	**6 points**
4.	GPA and Standing calculation:	**2 points**
5.	Code quality (readable, comment, reflection)	**2 points**
6.	Followed submission guidelines	**5 points**
7.	***Extra Credit***	***5 points***

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Reminder: Copy-pasting from AI or online sources without understanding is plagiarism. You should be able to explain your code in person if asked.

Tasks

Write a program that collects one student's semester with multiple courses and prints a transcript-style report:

- For each course, the user enters:
  - ✓ courseName (string, allow spaces)
  - ✓ creditHours (int ≥ 0)
  - ✓ letterGrade (A, A-, B+, B, B-, C+, C, C-, D, F)
- The program maps the letter grade to grade points per credit (e.g., A=4.0, B+=3.3, …), computes **Total Points = creditHours × gradePoints**, and shows:
  - ✓ Per-course: Credits, GradePts, TotalPts
  - ✓ Totals and Semester GPA = (sum TotalPts) / (sum Credits)
  - ✓ Standing:
    - GPA ≥ 3.70 → Dean's List
    - 2.00 ≤ GPA < 3.70 → Good Standing
    - GPA < 2.00 → Probation
- **EXTRA CREDIT:** (Sorting) Print courses sorted by credits (desc) or by course name (asc).

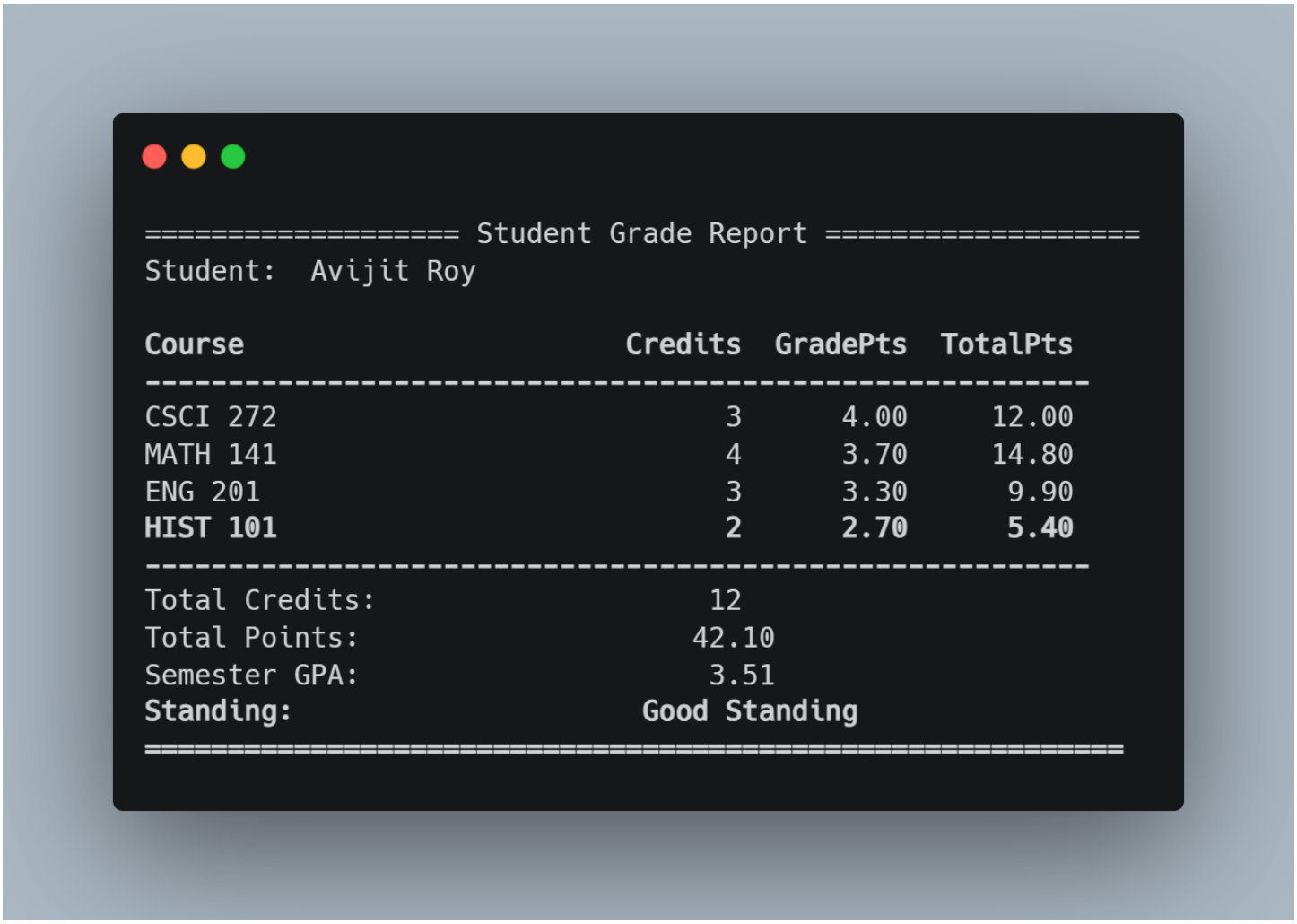
Input Rules

- Ask for student's full name (string, **getline**)
- Ask for number of courses N (1–10)
- For each course: name (string), credits (int ≥ 0), letter grade from the allowed set
- On bad input (e.g., credits not an int, invalid letter grade), re-prompt and recover using **cin.fail(), cin.clear(), cin.ignore(...).**

SEE NEXT PAGE FOR SAMPLE OUTPUT

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Sample Output



```
================== Student Grade Report ==================
Student:  Avijit Roy

Course                        Credits  GradePts  TotalPts
-----------------------------------------------------------
CSCI 272                            3      4.00     12.00
MATH 141                            4      3.70     14.80
ENG 201                             3      3.30      9.90
HIST 101                            2      2.70      5.40
-----------------------------------------------------------
Total Credits:                     12
Total Points:                   42.10
Semester GPA:                    3.51
Standing:                  Good Standing
===========================================================
```


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